

```

In[*]:= pivot[iStar_, jStar_, tableau_] := (
    newTableau = tableau;
    numberOfRows = Dimensions[tableau][[1]];
    numberCols = Dimensions[tableau][[2]];
    (* switch row tableau variables *)
    newTableau[[1, jStar]] = tableau[[iStar, numberCols]];
    newTableau[[iStar, numberCols]] = tableau[[1, jStar]];
    (* switch column tableau variables *)
    newTableau[[numberOfRows, jStar]] = -tableau[[iStar, 1]];
    newTableau[[iStar, 1]] = -tableau[[numberOfRows, jStar]];
    (* adjust loop bounds to skip 1st column and last row *)
    For[ii = 2, ii < numberOfRows, ii++,
        For[jj = 2, jj < numberCols, jj++,
            If[ii == iStar && jj == jStar,
                newTableau[[ii, jj]] = 1 / tableau[[iStar, jStar]];
            If[ii == iStar && jj != jStar,
                newTableau[[ii, jj]] = -tableau[[iStar, jj]] / tableau[[iStar, jStar]];
            If[ii != iStar && jj == jStar,
                newTableau[[ii, jj]] = tableau[[ii, jStar]] / tableau[[iStar, jStar]];
            If[ii != iStar && jj != jStar,
                newTableau[[ii, jj]] =
                    tableau[[ii, jj]] - tableau[[iStar, jj]] * tableau[[ii, jStar]] / tableau[[iStar, jStar]];
        ]
    ];
    Return[newTableau];
)

(* The given LP:

```

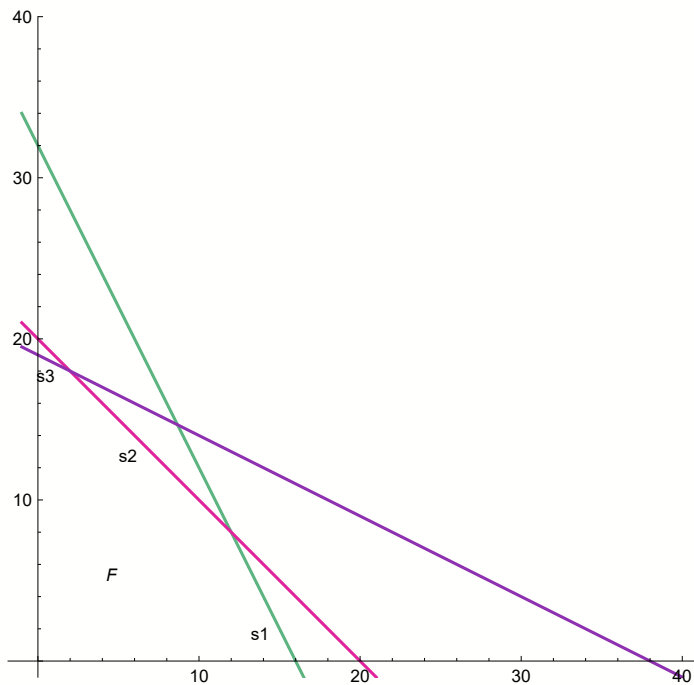
*)

```

In[ ]:= (* Part 1 graph feasible region. Shading is optional. *)
Clear[x1, x2]
z[x1_, x2_] := 17 x1 + 11 x2 - 9
g1[x1_, x2_] := 2 x1 + 1 x2
g2[x1_, x2_] := 1 x1 + 1 x2
g3[x1_, x2_] := 1 x1 + 2 x2
delz = {17, 11}; (* gradient of z *)
delg1 = {2, 1}; (* gradient of g1 *)
delg2 = {1, 1}; (* gradient of g2 *)
delg3 = {1, 2}; (* gradient of g3 *)
b1 = 32;
b2 = 20;
b3 = 38;
db1 = 0;
db2 = 0;
db3 = 0;
g1[x1_] := x2 /. Solve[g1[x1, x2] == b1, {x2}][[1, 1]]
g2[x1_] := x2 /. Solve[g2[x1, x2] == b2, {x2}][[1, 1]]
g3[x1_] := x2 /. Solve[g3[x1, x2] == b3, {x2}][[1, 1]]

Plot[{g1[x], g2[x], g3[x]}, {x, -1, 40},
  AspectRatio -> Automatic, PlotRange -> {-1, 40}, PlotHighlighting -> None ]

```



```

In[*]:= (* Part 2 *)
(* solve baseline tableau *)
db1 = 0;
db2 = 0;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b];
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

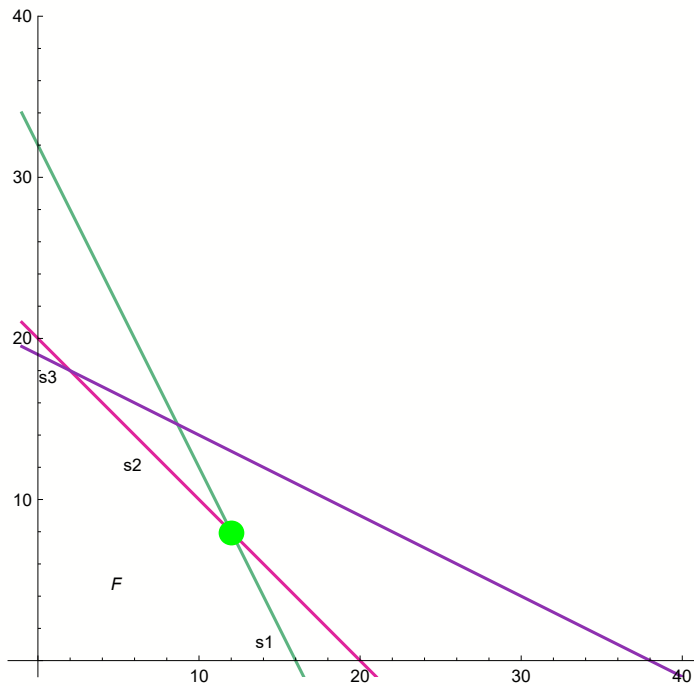
$$db = (\begin{matrix} 0 & 0 & 0 \end{matrix}) \text{ initial} = \begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 32 & 20 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

$$db = (\begin{matrix} 0 & 0 & 0 \end{matrix}) \text{ optimal} = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 12 & 8 & 10 & 283 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

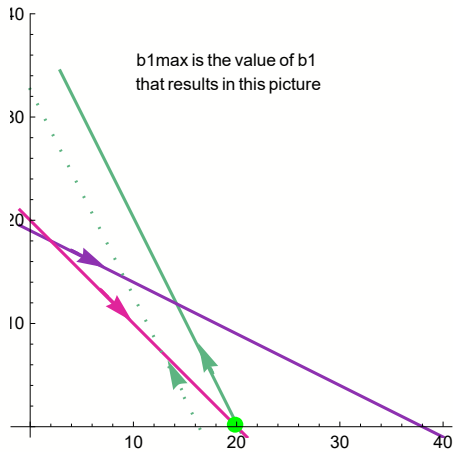
```

In[*]:= (* primal solution FOR THE BASELINE CASE before the Part 5 variations *)
zStarBase = 283;
x1StarBase = 12; x2StarBase = 8;
xStarBase = {x1StarBase, x2StarBase};
(* dual solution *)
wStar = 283;
y1Star = 6; y2Star = 5; y3Star = 0;
yStar = {y1Star, y2Star, y3Star};
(* for convenience define shorter variable names for yStar *)
y1 = y1Star; y2 = y2Star; y3 = y3Star;
y = {y1, y2, y3};

```



(* Part 3 *)



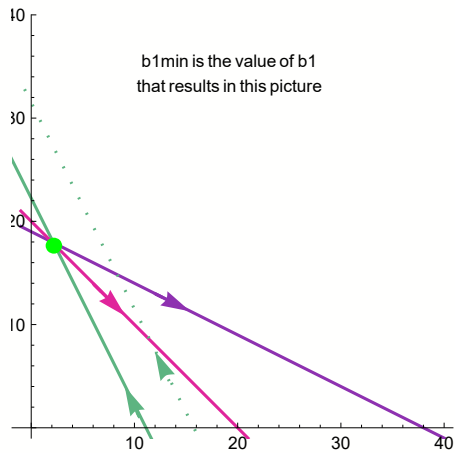
In[]:= **Solve[g2[u, v] == b2 && v == 0]**

Out[]:=
 $\{ \{ u \rightarrow 20, v \rightarrow 0 \} \}$

In[]:= **b1max = g1[20, 0]**
db1max = b1max - b1

Out[]:=
40

Out[]:=
8



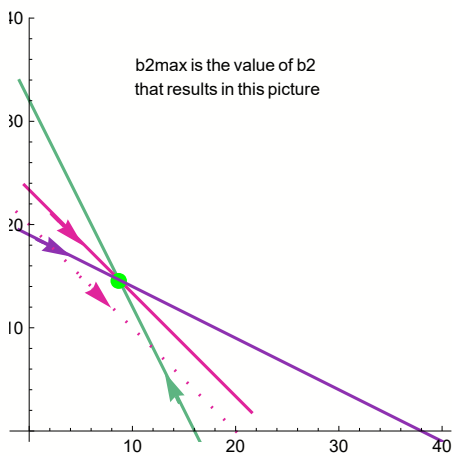
In[]:= **Solve[g2[u, v] == b2 && g3[u, v] == b3]**

Out[]:=
 $\{\{u \rightarrow 2, v \rightarrow 18\}\}$

In[]:= **b1min = g1[2, 18]**
db1min = b1min - b1

Out[]:=
 22

Out[]:=
 -10



In[]:= **Solve[g1[u, v] == b1 && g3[u, v] == b3]**

Out[]:=
 $\left\{\left\{u \rightarrow \frac{26}{3}, v \rightarrow \frac{44}{3}\right\}\right\}$

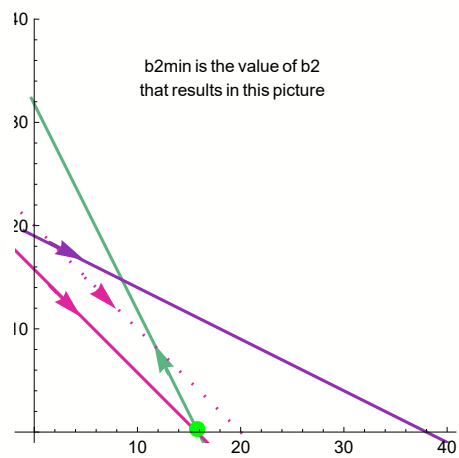
```
In[ ]:= b2max = g2[26 / 3, 44 / 3]
```

```
db2max = b2max - b2
```

```
Out[ ]:=
```

$$\frac{70}{3}$$

```
Out[ ]:=
```

$$\frac{10}{3}$$


```
In[ ]:= Solve[g1[u, v] == b1 && v == 0]
```

```
Out[ ]:=
```

```
{ {u -> 16, v -> 0} }
```

```
In[ ]:= b2min = g2[16, 0]
```

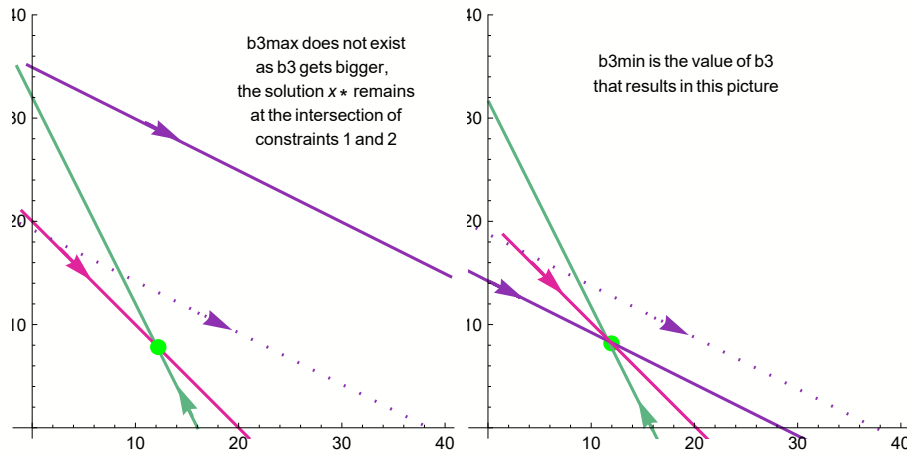
```
db2min = b2min - b2
```

```
Out[ ]:=
```

$$16$$

```
Out[ ]:=
```

$$-4$$



```
In[ ]:= Solve[g1[u, v] == b1 && g2[u, v] == b2]
```

```
Out[ ]:=  
{ {u -> 12, v -> 8} }
```

```
In[ ]:= b3min = g3[12, 8]  
db3min = b3min - b3
```

```
Out[ ]:=  
28
```

```
Out[ ]:=  
-10
```

(* Part 4 *)

(* db = db1 db2 db3

(* 4e to 4i follow using the min & max b values from part 3 *)

*)

```

In[*]:= (* 4.a *)
db1 = 0;
db2 = 2;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

db = ( 0 2 0 ) initial = 
$$\begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 32 & 22 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$


db = ( 0 2 0 ) optimal = 
$$\begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 10 & 12 & 4 & 293 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$


In[*]:= (* read new xStar & zStar from new optimal tableau and verify relations *)
Print["xStarNew = ", xStarNew = {10, 12}]
Print["zStarNew = ", zStarNew = 293]
Print["dxStar = ", dxStar = xStarNew - xStarBase]
Print["dzStar = ", dzStar = zStarNew - zStarBase]
dzStar == delz.dxStar == (y1Star delg1 + y2Star delg2 + y3Star delg3).dxStar ==
y1Star (delg1.dxStar) + y2Star (delg2.dxStar) + y3Star (delg3.dxStar) ==
y1Star db1 + y2Star db2 + y3Star db3

xStarNew = {10, 12}
zStarNew = 293
dxStar = {-2, 4}
dzStar = 10

Out[*]:=
True

```



```

In[*]:= (* 4.b *)
db1 = 3;
db2 = 0;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
(* 1st pivot *)
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = \begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 35 & 20 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

$$db = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 15 & 5 & 13 & 301 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

```

In[*]:= (* read new x & z from new optimal tableau and verify relations *)
Print["xStarNew = ", xStarNew = {15, 5}]
Print["zStarNew = ", zStarNew = 301]
Print["dxStar = ", dxStar = xStarNew - xStarBase]
Print["dzStar = ", dzStar = zStarNew - zStarBase]
dzStar == delz.dxStar == (y1Star delg1 + y2Star delg2 + y3Star delg3).dxStar ==
  y1Star (delg1.dxStar) + y2Star (delg2.dxStar) + y3Star (delg3.dxStar) ==
  y1Star db1 + y2Star db2 + y3Star db3

xStarNew = {15, 5}

zStarNew = 301

dxStar = {3, -3}

dzStar = 18

```

```

Out[*]=
True

```

```

In[*]:= (* 4.c *)
db1 = 3;
db2 = 2;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = (\begin{matrix} 3 & 2 & 0 \end{matrix}) \text{ initial} = \left(\begin{array}{cccccc} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 35 & 22 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

$$db = (\begin{matrix} 3 & 2 & 0 \end{matrix}) \text{ optimal} = \left(\begin{array}{cccccc} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 13 & 9 & 7 & 311 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

```

In[*]:= (* read new x & z from new optimal tableau and verify relations *)
Print["xStarNew = ", xStarNew = {13, 9}]
Print["zStarNew = ", zStarNew = 311]
Print["dxStar = ", dxStar = xStarNew - xStarBase]
Print["dzStar = ", dzStar = zStarNew - zStarBase]
dzStar == delz.dxStar == (y1Star delg1 + y2Star delg2 + y3Star delg3).dxStar ==
  y1Star (delg1.dxStar) + y2Star (delg2.dxStar) + y3Star (delg3.dxStar) ==
  y1Star db1 + y2Star db2 + y3Star db3
xStarNew = {13, 9}
zStarNew = 311
dxStar = {1, 1}
dzStar = 28

```

Out[*]=

True

```

In[*]:= (* 4.d *)
db1 = 3;
db2 = -2;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = (\begin{matrix} 3 & -2 & 0 \end{matrix}) \text{ initial} = \left(\begin{array}{cccccc} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 35 & 18 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

$$db = (\begin{matrix} 3 & -2 & 0 \end{matrix}) \text{ optimal} = \left(\begin{array}{cccccc} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 17 & 1 & 19 & 291 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

```

In[*]:= (* read new x & z from new optimal tableau and verify relations *)
Print["xStarNew = ", xStarNew = {17, 1}]
Print["zStarNew = ", zStarNew = 291]
Print["dxStar = ", dxStar = xStarNew - xStarBase]
Print["dzStar = ", dzStar = zStarNew - zStarBase]
dzStar == delz.dxStar == (y1Star delg1 + y2Star delg2 + y3Star delg3).dxStar ==
  y1Star (delg1.dxStar) + y2Star (delg2.dxStar) + y3Star (delg3.dxStar) ==
  y1Star db1 + y2Star db2 + y3Star db3
xStarNew = {17, 1}
zStarNew = 291
dxStar = {5, -7}
dzStar = 8

```

Out[*]=

True

(* 4e to 4i *)

(* See what happens when db1 = db1max or db1min, or when db2 = db2max or db2min, etc ... I will use 5.e (db1 = db2max) as an example. *)

```
In[*]:= (* 4.e db1 = db1max *)
db1 = db1max;
db2 = 0;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]
```

$$\text{db} = \begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 40 & 20 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

$$\text{db} = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 20 & 0 & 18 & 331 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

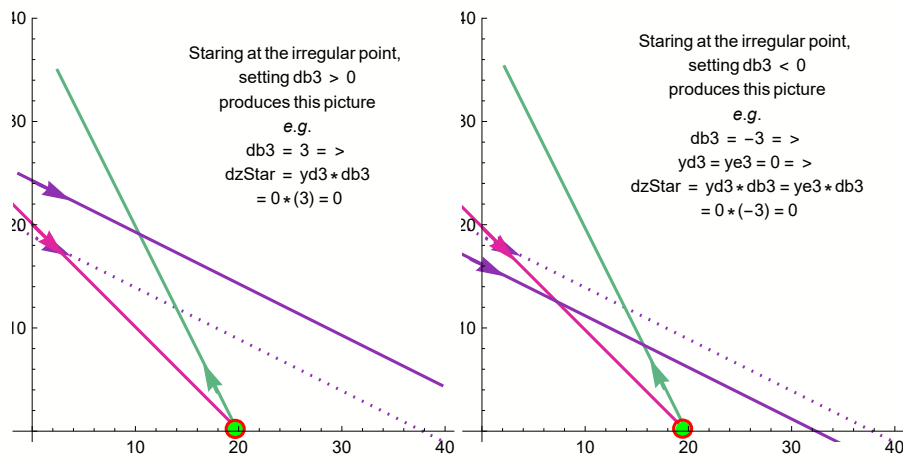
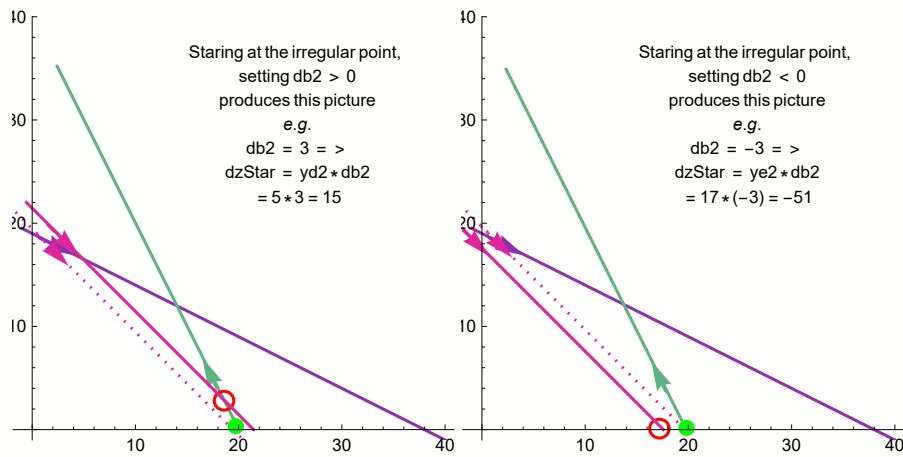
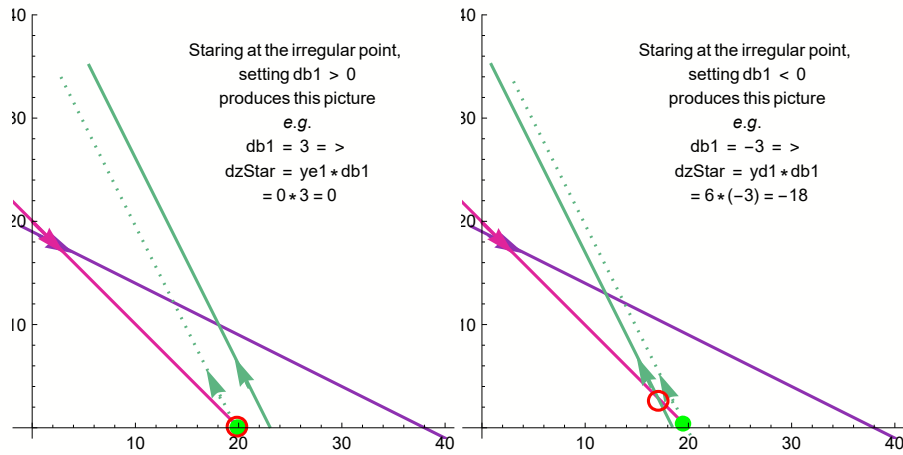
```
In[*]:= (* Notice that the optimal tableau has a 0 in the c-
row indicating a level set solution in the row-LP of the tableau i.e. the dual LP *)
```

```
In[*]:= (* Do an extra pivot to get the 2nd optimal tableau with a 2nd dual solution *)
e = pivot[3, 3, d];
MatrixForm[d];
MatrixForm[e];
Print["yd = ", MatrixForm[d], "ye = ", MatrixForm[e]]
```

$$y_d = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 20 & 0 & 18 & 331 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix} \quad y_e = \begin{pmatrix} & v1 & y1 & y3 & 1 & \\ s2 & 1 & -2 & -1 & 17 & y2 \\ x2 & 1 & -1 & 1 & 6 & v2 \\ -1 & 20 & 0 & 18 & 331 & w \rightarrow \min \\ & -x1 & -s1 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

(* THIS EXTRA ANALYSIS IS WHAT YOU DO FOR YOUR QUIZ:

Starting with the two optimal tableaux,
which both have the same primal solution of $x=(20,0)$ & $z=62$,
determine the rate of change dz^*/dbi for positive and negative dbi *)



(* THE QUIZ IS TO GET BOTH OPTIMAL TABLEAUX AND DO THE SAME SET OF PICTURES
AND COMPUTATIONS AT THE IRREGULAR POINT SOLUTIONS ARISING FROM f g h i

db=(db1min,0,0)

db=(0,db2max,0)

db=(0,db2min,0)

db=(0,0,db3min)

*)

In[*]:= (* 4.f db1 = db1min *)

db1 = db1min;

db2 = 0;

db3 = 0;

m0 = {" ", "y1", "y2", "y3", 1, " "};

m1 = {"x1", 2, 1, 1, -17, "v1"};

m2 = {"x2", 1, 1, 2, -11, "v2"};

mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};

mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};

a = {m0, m1, m2, mobj, mLast};

Print["db = (", db1, " ", db2, " ", db3, ") initial = ", MatrixForm[a]];

b = pivot[2, 2, a];

MatrixForm[b]; (* suppress output except for optimal tableau *)

c = pivot[2, 3, b];

MatrixForm[c];

d = pivot[3, 3, c];

MatrixForm[d];

Print["db = (", db1, " ", db2, " ", db3, ") optimal = ", MatrixForm[d]]

$$\text{db} = \begin{pmatrix} -10 & 0 & 0 \end{pmatrix} \text{ initial} = \begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 22 & 20 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

$$\text{db} = \begin{pmatrix} -10 & 0 & 0 \end{pmatrix} \text{ optimal} = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 2 & 18 & 0 & 223 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix}$$

In[*]:= e = pivot[2, 4, d];

Print["d=", MatrixForm[d], "e=", MatrixForm[e]]

$$d = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 2 & 18 & 0 & 223 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{pmatrix} e = \begin{pmatrix} & v1 & v2 & y2 & 1 & \\ s3 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & \frac{5}{3} & y3 \\ s1 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & \frac{23}{3} & y1 \\ -1 & 2 & 18 & 0 & 223 & w \rightarrow \min \\ & -x1 & -x2 & -s2 & -z \rightarrow \min & \end{pmatrix}$$

```

In[*]:= (* 4.g db2 = db2max *)
db1 = 0;
db2 = db2max;
db3 = 0;
m0 = {" ", "y1", "y2", "y3", 1, " "};
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = \left(0 \quad \frac{10}{3} \quad 0 \right) \text{ initial} = \begin{pmatrix} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 32 & \frac{70}{3} & 38 & -9 & w \rightarrow \min \\ -s1 & -s2 & -s3 & -z \rightarrow \min & & \end{pmatrix}$$

$$db = \left(0 \quad \frac{10}{3} \quad 0 \right) \text{ optimal} = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & \frac{26}{3} & \frac{44}{3} & 0 & \frac{899}{3} & w \rightarrow \min \\ -x1 & -x2 & -s3 & -z \rightarrow \min & & \end{pmatrix}$$

```

In[*]:= e = pivot[2, 4, d];
Print["d=", MatrixForm[d], "e=", MatrixForm[e]]

```

$$d = \begin{pmatrix} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 2 & 18 & 0 & 223 & w \rightarrow \min \\ -x1 & -x2 & -s3 & -z \rightarrow \min & & \end{pmatrix} e = \begin{pmatrix} & v1 & v2 & y2 & 1 & \\ s3 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & \frac{5}{3} & y3 \\ s1 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & \frac{23}{3} & y1 \\ -1 & 2 & 18 & 0 & 223 & w \rightarrow \min \\ -x1 & -x2 & -s2 & -z \rightarrow \min & & \end{pmatrix}$$

```

(* 4.h db2 = db2min *)
db1 = 0;
db2 = db2min;
db3 = 0;
m0 = { " ", "y1", "y2", "y3", 1, " " };
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = (\begin{matrix} 0 & -4 & 0 \end{matrix}) \text{ initial} = \left(\begin{array}{cccccc} & y1 & y2 & y3 & 1 & \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 32 & 16 & 38 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

$$db = (\begin{matrix} 0 & -4 & 0 \end{matrix}) \text{ optimal} = \left(\begin{array}{cccccc} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 16 & 0 & 22 & 263 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

```

In[ ]:= e = pivot[3, 3, d];
Print["d=", MatrixForm[d], "e=", MatrixForm[e]]

```

$$d = \left(\begin{array}{cccccc} & v1 & v2 & y3 & 1 & \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 16 & 0 & 22 & 263 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right) e = \left(\begin{array}{cccccc} & v1 & y1 & y3 & 1 & \\ s2 & 1 & -2 & -1 & 17 & y2 \\ x2 & 1 & -1 & 1 & 6 & v2 \\ -1 & 16 & 0 & 22 & 263 & w \rightarrow \min \\ & -x1 & -s1 & -s3 & -z \rightarrow \min & \end{array} \right)$$


```

(* 4.i db3 = db3min *)
db1 = 0;
db2 = 0;
db3 = db3min;
m0 = { " ", "y1", "y2", "y3", 1, " " };
m1 = {"x1", 2, 1, 1, -17, "v1"};
m2 = {"x2", 1, 1, 2, -11, "v2"};
mobj = {-1, b1 + db1, b2 + db2, b3 + db3, -9, "w→min"};
mLast = {"", "-s1", "-s2", "-s3", "-z→min", ""};
a = {m0, m1, m2, mobj, mLast};
Print["db = ( ", db1, " ", db2, " ", db3, " ) initial = ", MatrixForm[a]];
b = pivot[2, 2, a];
MatrixForm[b]; (* suppress output except for optimal tableau *)
c = pivot[2, 3, b];
MatrixForm[c];
d = pivot[3, 3, c];
MatrixForm[d];
Print["db = ( ", db1, " ", db2, " ", db3, " ) optimal = ", MatrixForm[d] ]

```

$$db = (\begin{matrix} 0 & 0 & -10 \end{matrix}) \text{ initial} = \left(\begin{array}{cccccc} & y1 & y2 & y3 & & 1 \\ x1 & 2 & 1 & 1 & -17 & v1 \\ x2 & 1 & 1 & 2 & -11 & v2 \\ -1 & 32 & 20 & 28 & -9 & w \rightarrow \min \\ & -s1 & -s2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

$$db = (\begin{matrix} 0 & 0 & -10 \end{matrix}) \text{ optimal} = \left(\begin{array}{cccccc} & v1 & v2 & y3 & & 1 \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 12 & 8 & 0 & 283 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right)$$

```

In[ ]:= e = pivot[2, 4, d];
Print["d=", MatrixForm[d], "e=", MatrixForm[e]]

```

$$d = \left(\begin{array}{cccccc} & v1 & v2 & y3 & & 1 \\ s2 & -1 & 2 & -3 & 5 & y2 \\ s1 & 1 & -1 & 1 & 6 & y1 \\ -1 & 12 & 8 & 0 & 283 & w \rightarrow \min \\ & -x1 & -x2 & -s3 & -z \rightarrow \min & \end{array} \right) e = \left(\begin{array}{cccccc} & v1 & v2 & y2 & & 1 \\ s3 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} & \frac{5}{3} & y3 \\ s1 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} & \frac{23}{3} & y1 \\ -1 & 12 & 8 & 0 & 283 & w \rightarrow \min \\ & -x1 & -x2 & -s2 & -z \rightarrow \min & \end{array} \right)$$